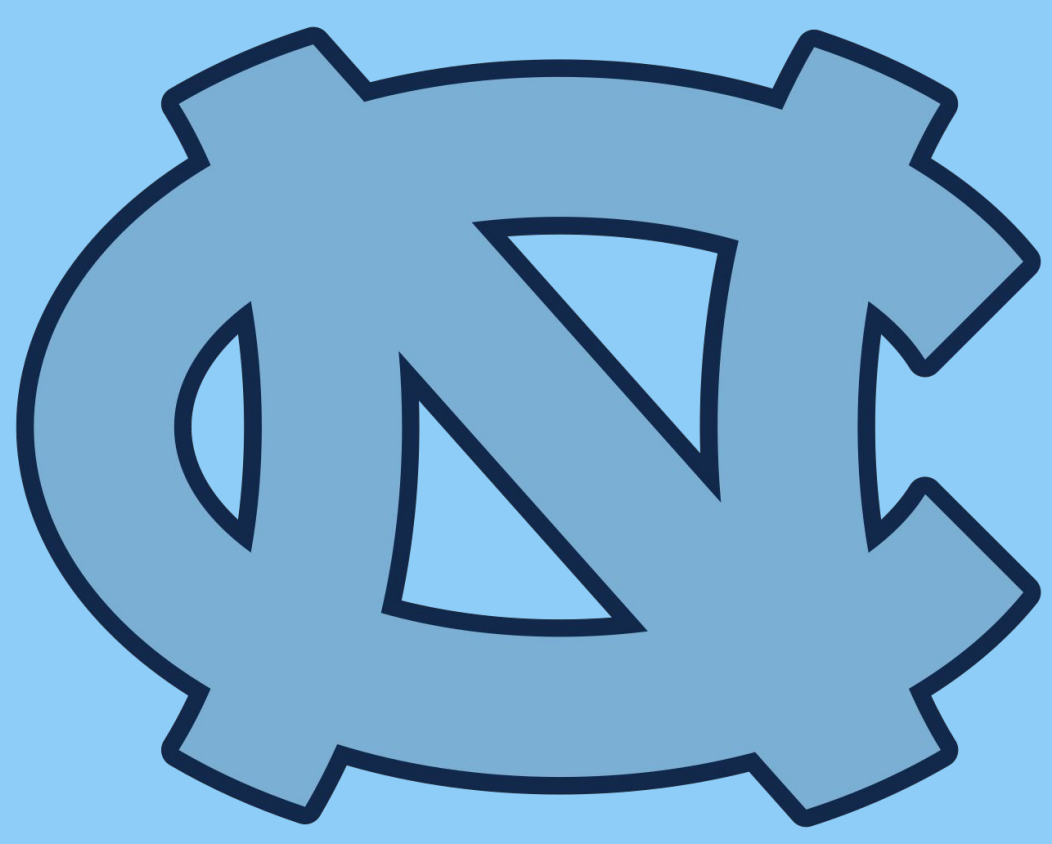
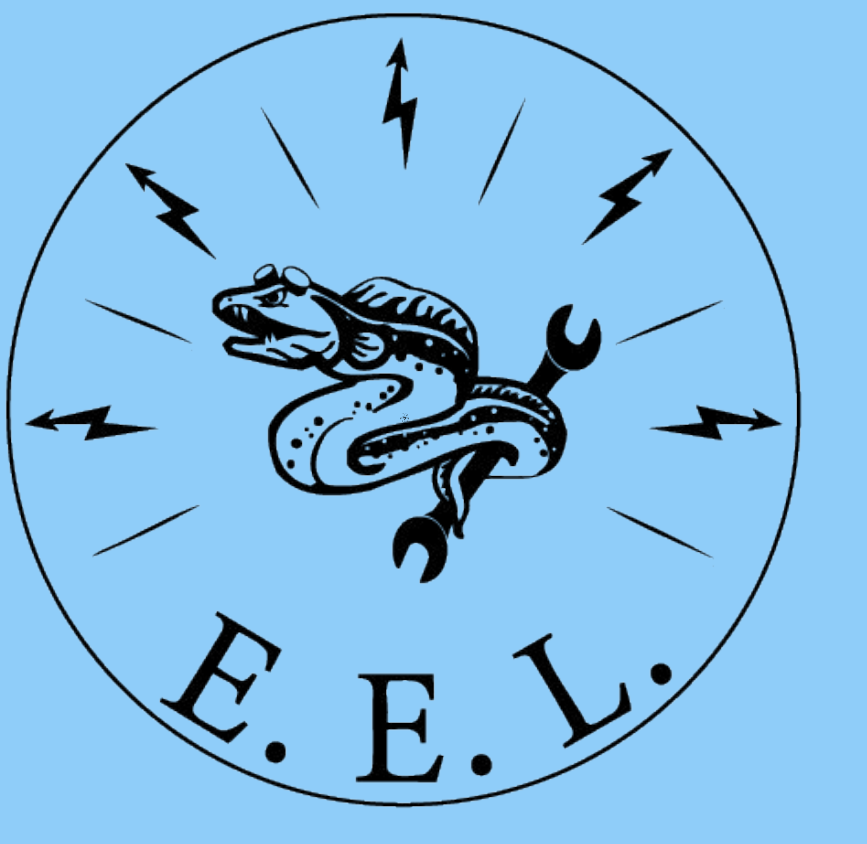




Experimental comparison of Halbach and circular magnetohydrodynamic drive configurations



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Importance

MHD propulsion has potential applications in silent marine propulsion, conductive-fluid pumping, and other systems where minimizing moving parts is beneficial. By comparing Halbach and circular drive configurations, this project explores how device geometry influences the performance and practicality of these technologies.

Background

Magnetohydrodynamic drives leverage a forcing effect on conductive media (in this case saltwater) by the Lorentz force. This force is defined as:

$$\mathbf{f} = \mathbf{J} \times \mathbf{B}$$

The cross product where $\mathbf{B}$ is the magnetic field and $\mathbf{J}$ is the current density is defined as:

$$\mathbf{J} = \sigma \mathbf{E} = \rho_c \mathbf{v}_d$$

where σ is the electrical conductivity of the fluid and $\mathbf{E}$ is the applied electric field. This can also be expressed with ρ_c as the charge density of the moving carriers and $\mathbf{v}_d$ as their drift velocity between the charged plates.

The MHD force is strongest when the current density and magnetic field are both large and remain nearly perpendicular, since the force magnitude scales as

$$|\mathbf{f}| = JB \sin(\theta)$$

In a standard MHD drive (Figure 1), edge effects can reduce performance. The magnetic fields spread outward, causing $\mathbf{J}$ and $\mathbf{B}$ to become less nearly perpendicular (Figure 2). This reduces the effective Lorentz force on the water and lowers propulsion efficiency. As a result, drive performance can be improved while also increasing magnetic-field concentration and current/field alignment within the active flow region.

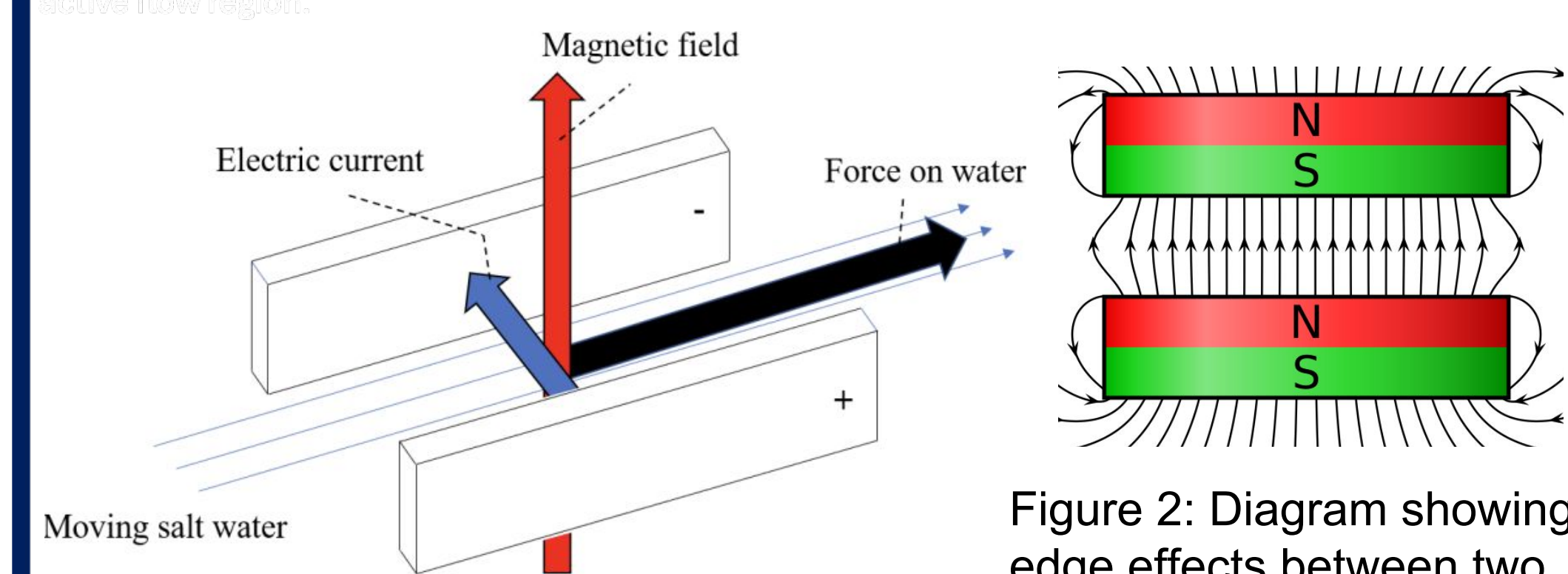
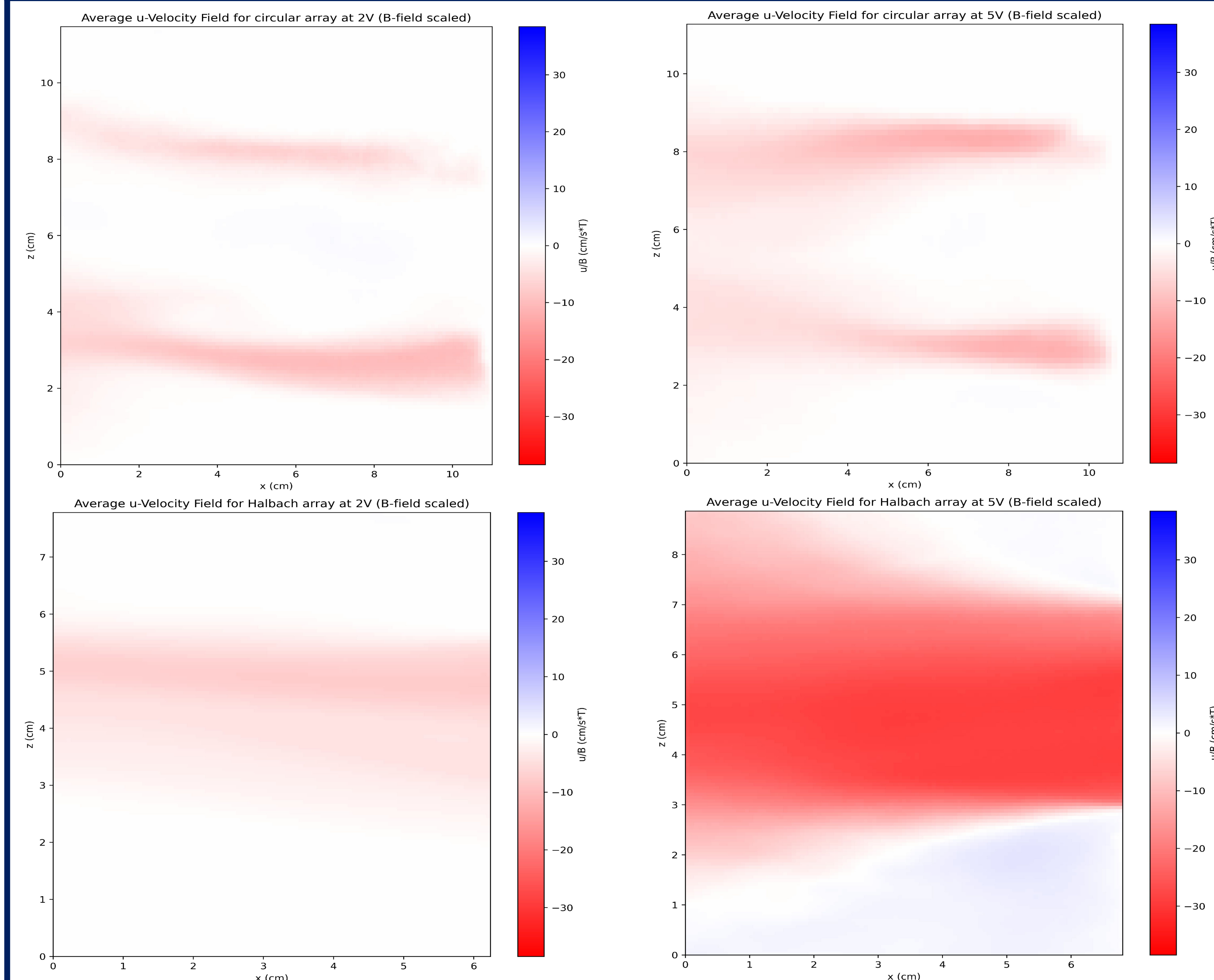


Figure 1: Diagram of a standard MHD drive

Figure 2: Diagram showing edge effects between two magnets

Results



Figures 6-9: Average velocities normalized by B field. Top Left: Circular at 2V. Top Right: Circular at 5V. Bottom Left: Halbach at 2V. Bottom Right: Halbach at 5V.

Efficiency was calculated using the property of back-emf, where a moving conductor creates an opposing voltage. In this case, the conductor is salt water. Looking next at efficiency, the total current is:

$$I = \frac{V}{R} - \frac{E_l w}{R}$$

where I is the current, V is the applied voltage, R is the resistance of the salt water, E_l is the back emf generated by the moving conductor, w is the distance between the electrodes, and $|\vec{v}|$ is the velocity of the water. Defining the efficiency, η , as the ratio of output to input,

$$\eta \equiv \frac{E_l w}{V} = \frac{B |\vec{v}| w}{V} \quad (\text{To compare the geometry independent of field strength, we report a B-normalized efficiency metric})$$

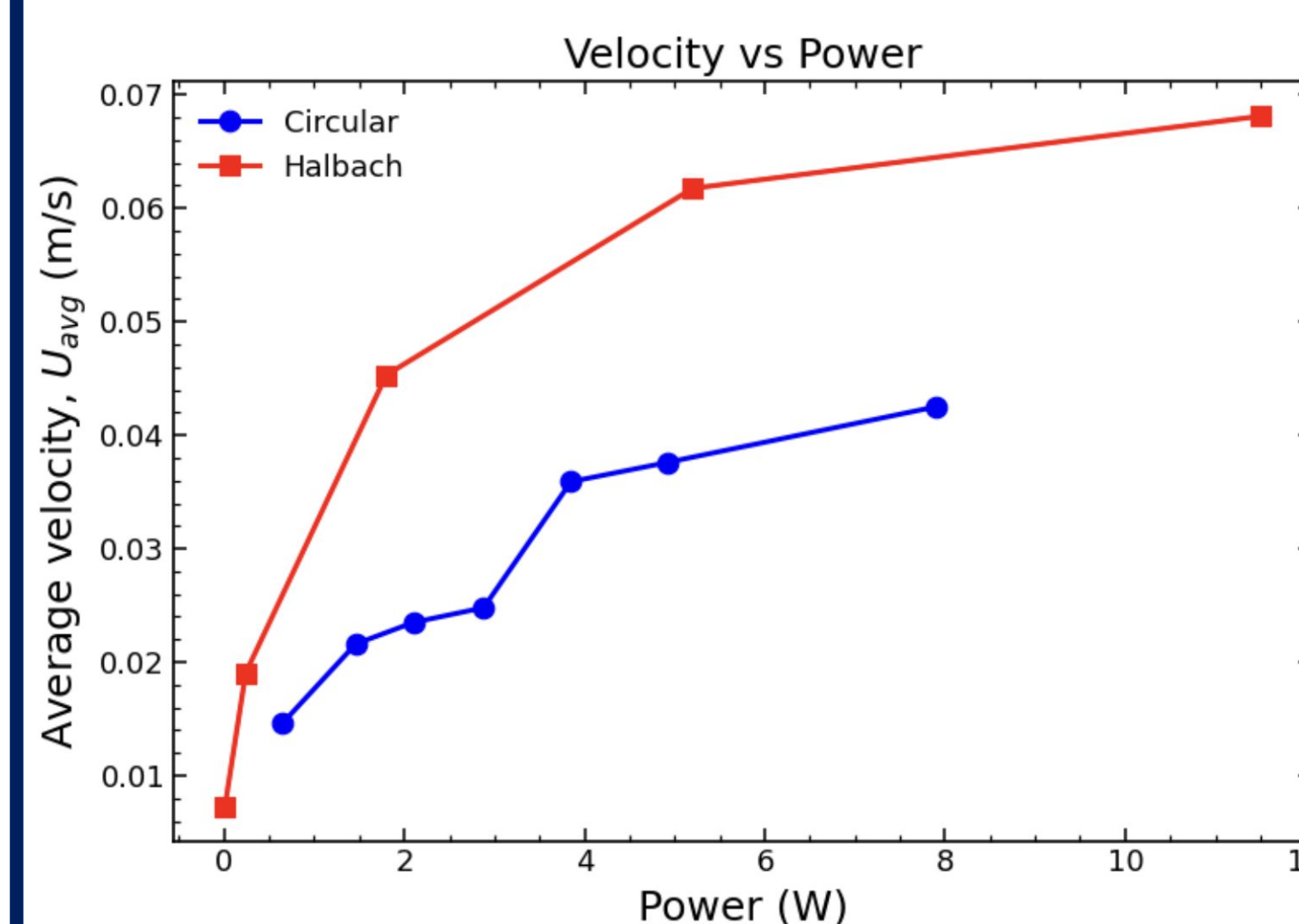


Figure 10: Graph of velocity versus input power (error bars are <1% and not visible)

- The Halbach performs better in a pure exhaust velocity test.

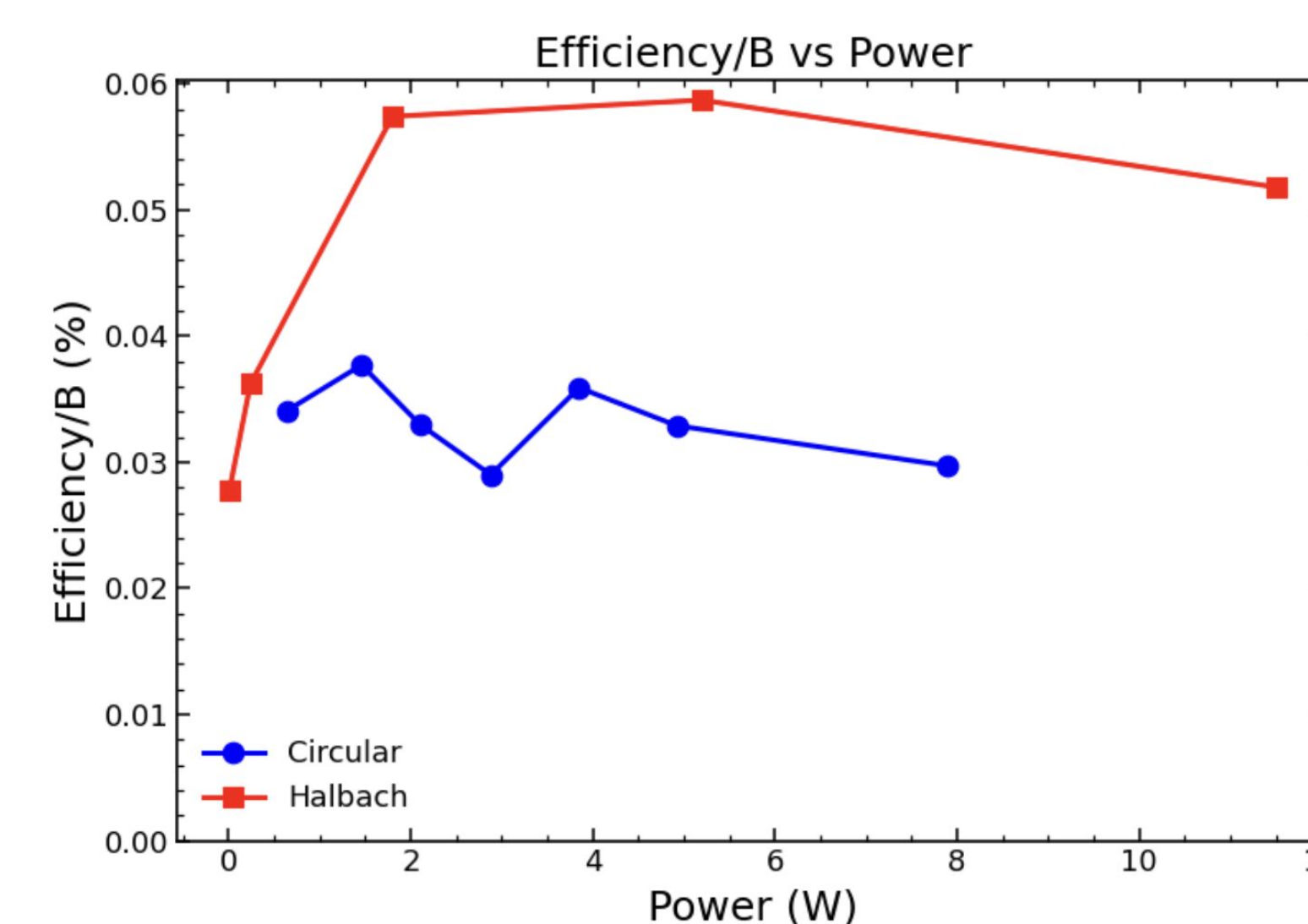


Figure 11: Graph of Efficiency/B versus input power (error bars are <1% and not visible)

- The Halbach has a normalized efficiency **approximately 50% greater** than the circular orientation under similar power levels

Drive design and experimental setup

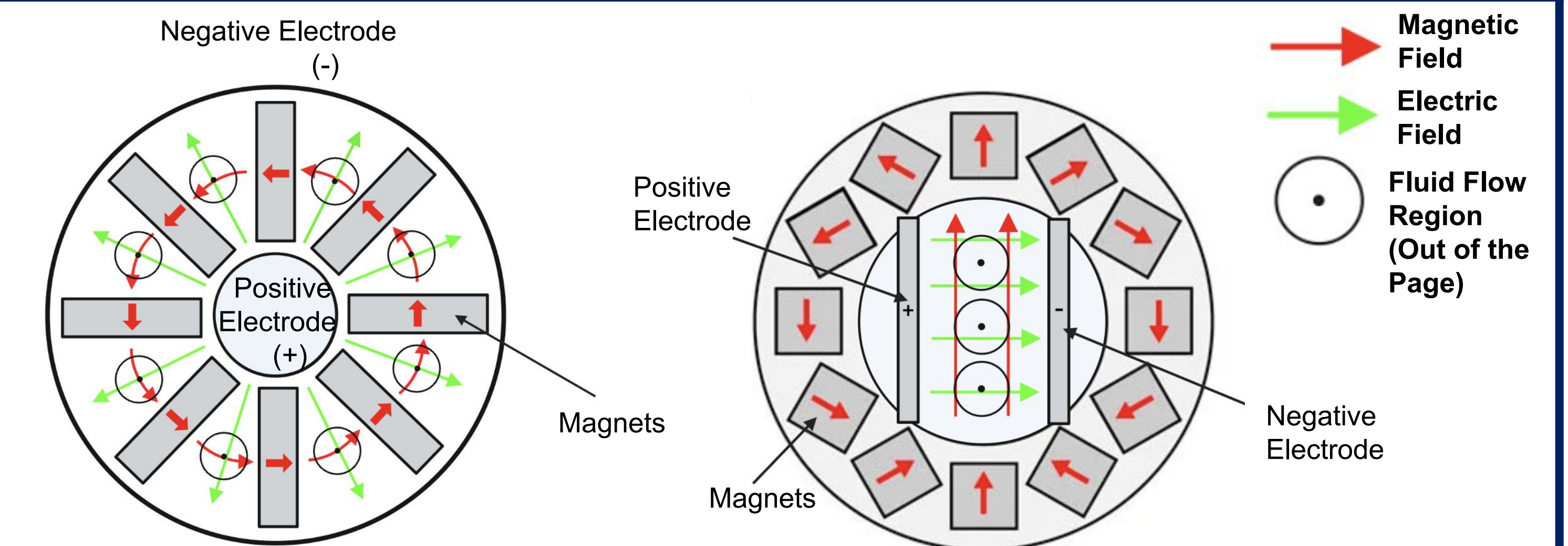


Figure 3: Front facing diagram of the circular oriented drive configuration

Electrode material: copper-nickel marine alloy

Electrode gap: 3.5 cm

Effective area: 25.56 cm²

Magnet grade: N52

Field strength: 0.40 T

To test the different configurations possibly leading to better uniformity, the circular and Halbach drive designs were created.

Figure 4: Front facing diagram of the Halbach drive configuration

Electrode material: 304 stainless steel

Electrode gap: 3.8 cm

Effective area: 17.34 cm²

Magnet grade: N45

Field strength: 0.26 T

- The circular drive configuration was investigated as an alternative geometry expected to produce a compact design possibly leading to fewer edge effects and better magnetic uniformity than a standard MHD.
- The Halbach configuration was selected because the geometry itself is designed to concentrate the magnetic field and more uniformly distribute it within the bore, or active flow region.

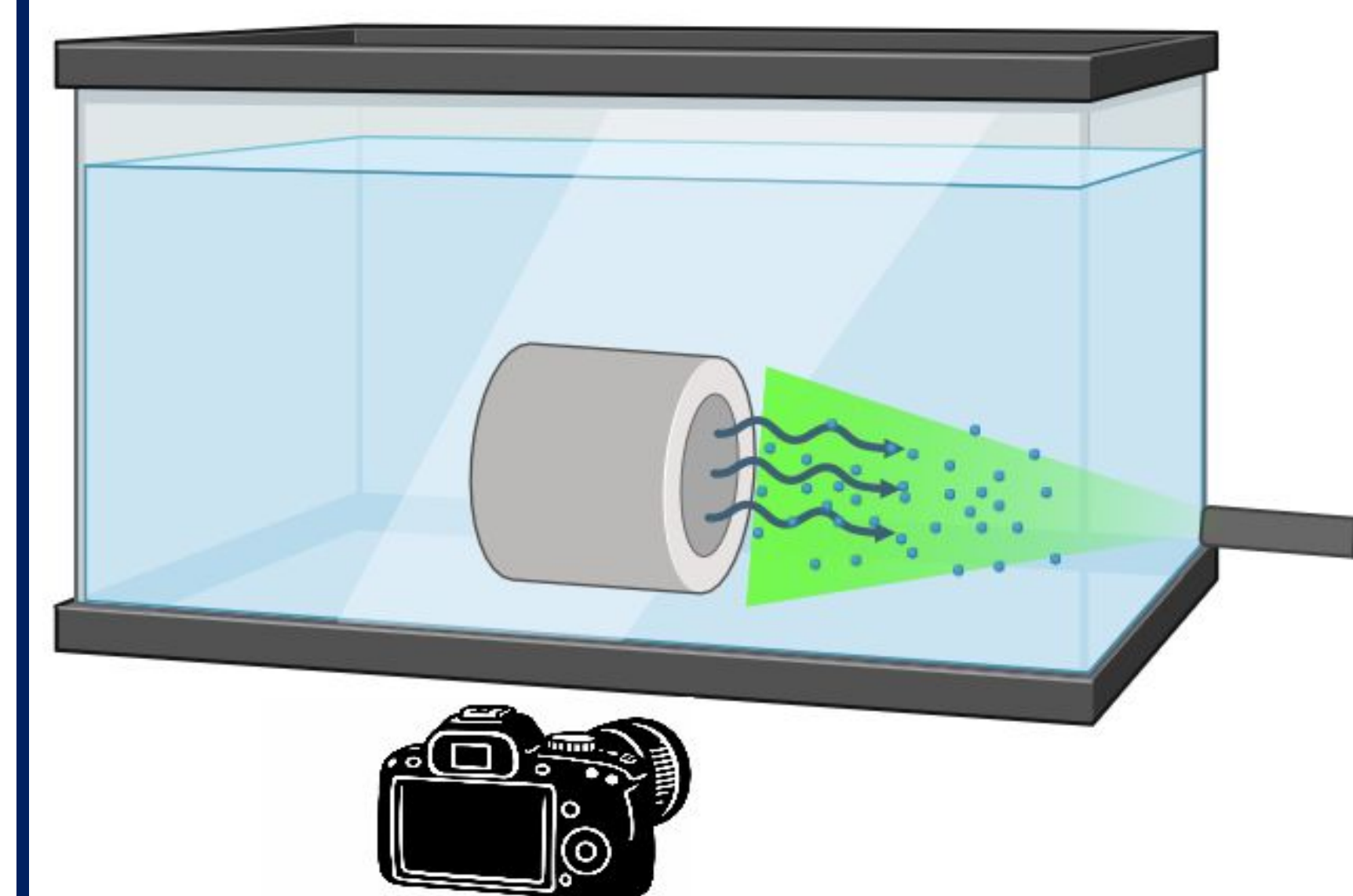


Figure 5: Tank, Camera, laser, and PIV setup

- Both drives were tested in a salt-water tank prepared to approximate seawater conditions. The fluid had a density of approximately 1.028 g/mL (1028 kg/m³), giving an extrapolated conductivity of about 5.5 S/m.

- Flow exiting the drives was recorded using 4K video at 120 fps, and the jet motion was analyzed from a "laser sheet" cross section of the drive, illuminating the PIV particles in the flow region.
- PIVlab was used to extrapolate flow velocity. These measurements were then used to compare exhaust velocity, power input, and propulsion efficiency between the two configurations.

Acknowledgments/References

- [1] S. H. Brown, J. S. Walker, N. A. Sondergaard, P. J. Reilly, and D. E. Bagley, "Propulsive Efficiencies of Magnetohydrodynamic Submerged Vehicular Propulsors," David Taylor Research Center, DTRC-90/009, Apr. 1990.
- [2] N. Schmelzer, "Magnetohydrodynamic Salt Water Drive," College of Saint Benedict and Saint John's University, 2019.
- [3] "Halbach array," Wikipedia, The Free Encyclopedia. Accessed Apr. 19, 2026.
- [4] Geek3, "VFPT flat magnets gap.svg," Wikimedia Commons, Dec. 16, 2018. Licensed under CC BY-SA 4.0.

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Conclusion

- The Halbach configuration produced higher exhaust velocities across the tested power range and showed an **approximately 50% greater normalized efficiency** than the circular design. These results support the idea that improved magnetic field concentration and uniformity in the active flow region can significantly improve MHD drive performance.
- Future work should further isolate geometry by matching electrode materials and dimensions more closely and testing the combination of Halbach geometry with stronger magnets. It would also be beneficial to test other methods of estimating efficiency for MHD propulsion systems.